

Fermium Hazard Mapper: an open pipeline for flood and landslide risk mapping, trained and tested on radar-observed floods

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Abstract

Flood susceptibility maps for Bangladesh are usually trained on a few hundred surveyed points and scored on a random share of the same points, so their accuracy says little about unmapped ground. Fermium Hazard Mapper is an open Python pipeline and web application that trains flood models on extents mapped from Sentinel-1 radar, scores every model on whole areas held out from training, and publishes asset-level and grid-level risk with the evidence behind it. One configuration file per region drives a chain of command-line stages from data download to calibrated scores, composite risk, administrative summaries and a validation suite that compares models, label sources and time periods on identical spatial blocks. A national partition of Bangladesh into 8 flood regions, each holding one flood regime, covers all 61 lowland districts and 114,569 mapped assets, and a landslide region covers the Chittagong Hill Tracts. The results are served openly at <https://fermium.systems/hazmapper>, from the division down to the union, and a suite of 159 tests guards each defect found in an audit of the first release.

Keywords: flood susceptibility, Sentinel-1, spatial cross-validation, risk mapping, open-source software, Bangladesh

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Code metadata

Nr.	Code metadata description	Metadata
C1	Current code version	v3.0
C2	Permanent link to code/repository used for this code version	https://github.com/rakibhhriday/ri-mes-mapathon
C3	Permanent link to reproducible capsule	Archived release: [software DOI, assigned on release]
C4	Legal code license	MIT
C5	Code versioning system used	git
C6	Software code languages, tools and services used	Python 3.11; GeoPandas, Rasterio, GDAL, scikit-learn, PyTorch Geometric, PyKrig, PySAL; Google Earth Engine; FastAPI, SQLite, PMTiles and MapLibre GL
C7	Compilation requirements, operating environments and dependencies	Linux or macOS; dependencies pinned in requirements.txt ; a Google Earth Engine project for the Sentinel-1 stage
C8	Link to developer documentation/manual	https://github.com/rakibhhriday/ri-mes-mapathon#readme
C9	Support email for questions	rakibhhriday@fermium.systems

1. Motivation and significance

Bangladesh floods every monsoon, yet the susceptibility maps published for its flood regions rarely show how they would fare on ground they were not trained on. Studies in the Teesta basin, the Sylhet haor and the south-west coast report AUCs between 0.81 and 0.984 [1, 2, 3, 4], each from a random split of a few hundred to a few thousand points. A random split of spatially clustered points tests a model mostly on locations next to its training points, and block cross-validation, which holds out whole areas, exists to remove that advantage [5]. Where the two have been compared the gap can be large, as in a forest-biomass model that explained more than half the variance under random validation and almost none under spatial validation [6]. The flood locations themselves vary in kind, from field surveys

to single satellite scenes, so the same accuracy figure can measure agreement with quite different things.

A stricter design is now practical, because Sentinel-1 radar sees through monsoon cloud and has been processed into flood maps of high accuracy, including for the 2017 Bangladesh floods [7, 8], while OpenStreetMap records the roads, schools, hospitals, shelters and embankments that a flood would reach. What has been missing is software that joins them in one reproducible chain, trains on what the radar saw rather than on terrain thresholds, holds out whole areas when scoring, and puts the result in front of planners with its limits attached.

Fermium Hazard Mapper fills that gap by answering three questions for a configured region, namely where the ground is likely to flood, which mapped assets and which people stand on that ground, and how far those people are from the services that would help them cope. It then reports how well each answer holds up on areas held out from training, on floods that occurred after the training period, and against a model-free reference. This article describes that software, the evidence it reports for Bangladesh, and how it can be pointed at a new region.

2. Software description

2.1. Software architecture

The system has three layers, a processing pipeline, a validation suite and two front ends, and they communicate only through files (Fig. 1). The pipeline is a Python package driven by one command-line interface, in which each stage reads the files the previous stage wrote into the region’s data directory and writes its own. A region runs end to end with a single command,

```
python -m pipeline.cli -c configs/national/central.yaml run
```

and any stage can be rerun alone after a change to its settings. Because stages share nothing but files, every intermediate product can be opened in a GIS for inspection.

A region is defined by a YAML file that states only what differs from a shared base configuration and names that base with an `extends` key. The base holds the model settings, weights and thresholds, so a change to any of them reaches every region and results stay comparable. A region file carries its extent, either a bounding box or a list of districts, together with a country code, a projected coordinate system, data directories and, for validation, the

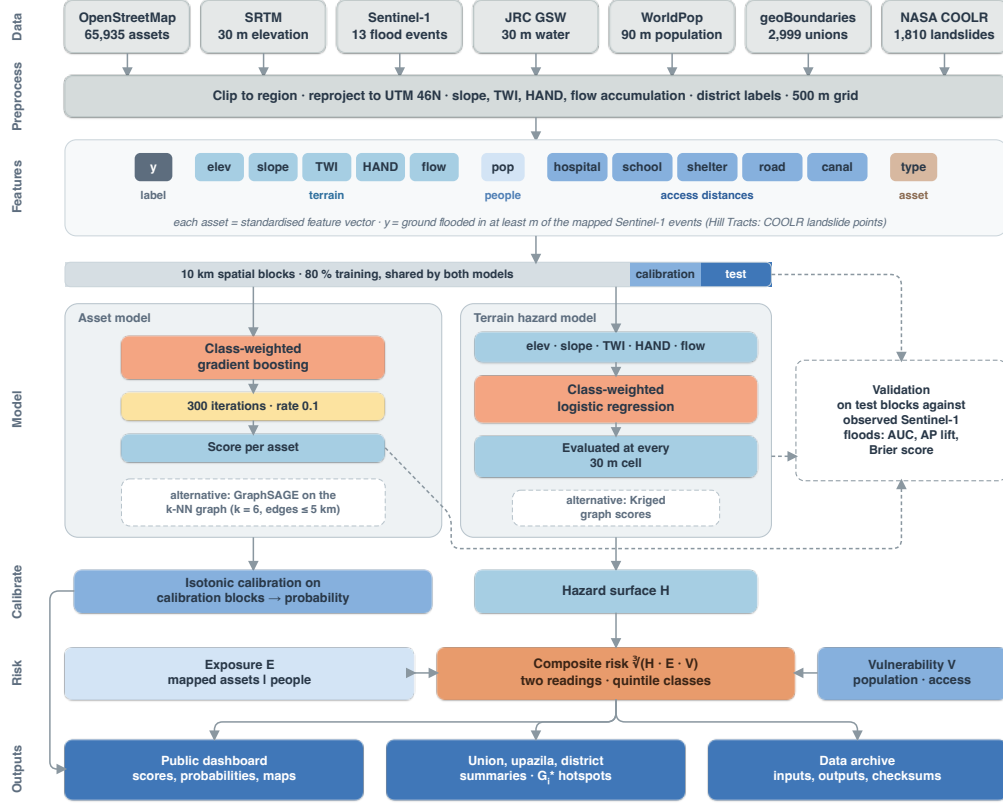


Figure 1: The processing chain, top to bottom. Each asset enters the models as a vector of terrain, population, access-distance and asset-type features, with a label from the Sentinel-1 flood frequency. Every model trains on the same spatial blocks, and dashed connectors mark what the validation scores on the test blocks. In the landslide region a model fitted to a mapped inventory takes the place of the flood labels.

flood events to map from radar, each with the dates of its window and the report those dates came from. A region defined by districts keeps only the assets, grid cells and validation points inside them, so the regions of a national partition never count the same asset twice.

The public site is the web application, which serves the outputs from an SQLite database and PMTiles vector tiles through a FastAPI service and a MapLibre GL client, so that every asset of every region is drawn at once and panning and filtering happen in the browser (Fig. 2). The build step also places every asset in its union, upazila, district and division, and sums each region’s results up that hierarchy, so the map shows one administrative level at a time, chosen by zoom, and a click or a cascade of pickers narrows the assets and counters to one area. The official return-period scenarios of the government’s Multi-hazard Risk and Vulnerability Assessment can be laid under the assets, drawn live from the server that publishes them, and the address of the page records the view, so a copied link or a downloaded image reproduces it. A Streamlit dashboard that reads the outputs directly remains for local use. Neither front end computes anything, and every value a visitor sees can therefore be traced to a file the pipeline wrote.

2.2. *Software functionalities*

The data stages listed in Table 1 read OpenStreetMap assets either from the Overpass API, in tiles that are cached and split when the server cannot answer, or from a national extract converted once into an indexed file. Assets are clipped to the region and to the country, so that an extent crossing a land border keeps only the home side. Slope, the topographic wetness index, height above nearest drainage and flow accumulation come from 30 m SRTM elevation, population from WorldPop, and administrative units from geoBoundaries, with divisions dissolved from their districts. The Sentinel-1 stage maps each configured event in Google Earth Engine by change detection against a dry-season baseline, following the UN-SPIDER practice [9], and splits a download the service refuses into quarters. It removes permanent water with the JRC surface-water record [10], and averages the event masks into a flood-frequency layer that becomes the training label.

The asset model is set in the configuration, with class-weighted gradient boosting as the default and a random forest, logistic regression and a two-layer GraphSAGE network [11] on a k -nearest-neighbour graph of assets available as alternatives. Scores are calibrated by isotonic regression [12] fitted on blocks the model never trained on, with each calibration level shrunk

Stage	What it does
download, ingest	fetch assets, elevation, population, boundaries and surface water, and clip them to the region
clip	drop assets across a land border from an asset file fetched earlier
preprocess	derive terrain rasters and write both label sources, radar-observed and terrain-threshold
sentinel1	map each configured flood event from radar and build the flood-frequency layer
features, graph	sample features at every asset and assign assets to 10 km spatial blocks
train	fit the configured asset model on training blocks and calibrate it on calibration blocks
krige, risk	build the terrain and Kriged hazard surfaces, exposure, vulnerability and composite risk, and summarise them by administrative unit
landslide	fit a landslide susceptibility model to a mapped inventory
validate, crosscheck	score models and surfaces against observed floods, and radar masks against an independent flood database
benchmark, sensitivity	compare models and label sources over repeated block assignments, run the temporal hold-out, and perturb the risk weights
run	run the whole chain for one region

Table 1: Pipeline stages, each a subcommand of the command-line interface.

towards the middle in proportion to how few points it holds, so that a level of three flooded points cannot return a certainty. Two hazard surfaces are written, a logistic model of each grid cell’s own terrain and an ordinary Kriging of the asset scores, and the composite risk uses the one the configuration names. Composite risk is the geometric mean of hazard, exposure and vulnerability on a 500 m grid, the aggregation the INFORM index uses [13], and it is written twice, once for mapped infrastructure and once for population, with hotspots located by the Getis–Ord G_i^* statistic [14]. For the landslide region the pipeline fits a logistic model to a mapped inventory, drawing background points only from the ground the inventory covers so that the model learns terrain rather than the mapping footprint [15].

The validation suite is what distinguishes the system from a mapping tool. Every model is scored on 10 km blocks held out from training, and the **benchmark** stage refits it over 20 block assignments so that each difference is reported with its spread. On identical features, assets and blocks, the same stage compares the configured models with one another and with the wetness index alone, compares training on radar labels with training on terrain-threshold labels, and trains on earlier floods to score against the latest year. The **crosscheck** stage compares the radar masks with the MODIS-based Global Flood Database [16], and the **sensitivity** stage recomputes the composite under alternative and randomly perturbed weights.

2.3. Safeguards against silent failure

The first release of the system contained a coordinate-system error that sampled rasters stored in UTM with longitude and latitude, so every terrain value came back as the nodata fill and every training label was zero. Nothing failed, and the model it produced was published with a validation score. The current release stops on inputs of that kind, halting if any feature comes out constant, if the labels hold a single class, if a radar event returns no imagery, or if a zonal summary overlaps no unit. Units without mapped assets are left unscored instead of scored zero, and a missing output is reported as missing.

The audit found seventeen defects and a later review an eighteenth, assets across the border in India, and each one that was a computation is now guarded by a test in a suite of 159, which also fails if the dashboard reintroduces mock data or random values. No figure in the documentation is typed by hand, because a build script reads each region’s outputs and writes one LaTeX macro per value. A value the pipeline has not produced renders as a visible marker, and a test checks that every macro used in the documentation exists. Rerunning the pipeline and rebuilding a document therefore updates its text and its results together.

3. Illustrative example

The pipeline was developed on three bounding-box regions, Rangpur and Rajshahi for riverine flooding, Sylhet for flash flooding and the south-west coast for cyclone surge, and then pointed at the whole country. The national partition assigns each of the 61 lowland districts to one of 8 regions by flood regime, and each region needed only a configuration file listing its districts and its flood events, every event checked against a situation report and

Region	Assets	Flood-prone (%)	AUC, held-out blocks	AUC vs observed	Radar-label gain
North-west rivers	21,893	6.2	0.821 ± 0.049	0.878	+0.103 (19/20)
Jamuna east bank	5,415	6.4	0.815 ± 0.074	0.726	+0.156 (19/20)
Sylhet haor	13,402	14.9	0.835 ± 0.049	0.840	+0.079 (19/20)
Central	20,218	3.0	0.914 ± 0.042	0.939	+0.118 (19/20)
West-central	4,398	3.2	0.724 ± 0.186	0.574	+0.024 (12/20)
South-west coast	26,039	5.0	0.799 ± 0.058	0.802	+0.070 (16/20)
Eastern plains	9,461	4.3	0.743 ± 0.122	0.854	+0.181 (17/20)
Chattogram coast	13,743	3.5	0.832 ± 0.118	0.637	+0.084 (15/19)

Table 2: The national partition. AUC on held-out 10 km blocks is the gradient-boosted model’s mean $\pm$ standard deviation over 20 block assignments; AUC against observed floods is scored on the held-out blocks of one assignment; the radar-label gain is the AUC won by training on radar-observed flooding instead of terrain-threshold labels, with the number of assignments in which radar labels came out ahead.

against Sentinel-1 coverage of its window before the run. The same commands scored 114,569 mapped assets (Table 2), and the web build placed them in the national administrative hierarchy (Fig. 2).

The table carries the evidence a user needs before trusting a region. Training on radar-observed flooding beats training on terrain thresholds in every region, but in West-central, whose two waterlogging events were documented only in newspapers, by a margin within noise, and its skill varies widely between block assignments, so the dashboard marks it as low-data. The graph network the system was first built around trails the best ordinary model in every region, by 0.005 to 0.088 in AUC, which is why gradient boosting is the default, and the comparison stays in the pipeline so that a future model has to beat it on the same blocks.

The same run reports what a user should not take from the maps. In the riverine validation regions, a model trained only on floods up to 2022 ranks the ground the 2024 floods reached at 0.801 and 0.797, but the radar record of earlier flooding ranks it better, at 0.896 and 0.851. On the south-west coast both hazard surfaces are close to chance across the grid, at 0.542 for the terrain surface, because surge flooding follows embankments and tides that terrain does not describe. The dashboard states each of these limits beside the map it qualifies, and the landslide model, at an AUC of 0.729 from 1,810 landslides mapped after a single storm, carries its single-event caveat above the map.

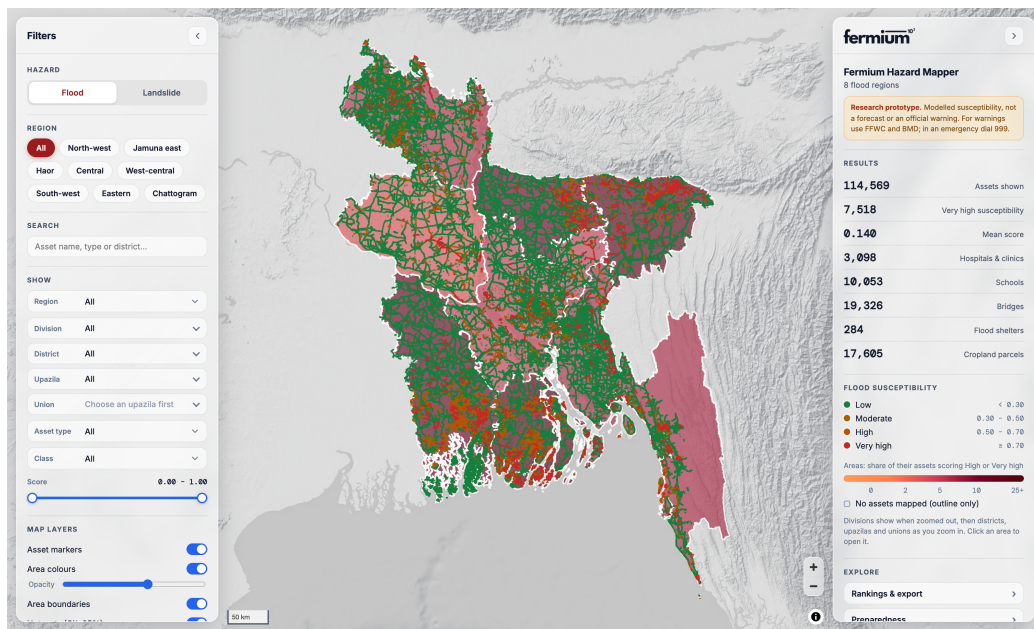


Figure 2: The web application in its national flood view. Asset markers are coloured by modelled susceptibility class and divisions by the share of their assets in the High or Very high class; zooming in replaces divisions with districts, upazilas and unions. Filters, administrative pickers and layers sit on the left; counters for whatever the filters leave visible, the legend and the model figures sit on the right.

4. Impact

The system changes what a flood susceptibility result for Bangladesh can be checked against. Its comparisons of models, label sources and time periods run on identical blocks and are repeated over many block assignments, so a claim such as “the graph network helps” or “radar labels help” becomes a number with a spread that another researcher can reproduce with one command. That design is what showed the graph architecture adding nothing over a gradient-boosted tree, and what showed the flood record outranking the model on the next riverine floods. A new model enters the same comparisons once it is added to the asset-model module as another configurable type.

For planners and responders, the public dashboard at <https://fermium.systems/hazmapper> gives division-, district-, upazila- and union-level figures for the whole country without a sign-in, beside the official scenario maps, and a link or an image of any view can be passed on. Every figure on it can be traced to a pipeline output, and units without mapped assets are shown as missing rather than as safe, which matters in rural areas where OpenStreetMap is thin. The dashboard names the Flood Forecasting and Warning Centre and the Bangladesh Meteorological Department as the sources for warnings, since the maps describe typical monsoon conditions and not a forecast.

The pipeline is not tied to Bangladesh’s regions, since a new region needs only a configuration file with a bounding box or a list of districts, a country code, a projected coordinate system and a list of flood events with their dates. The same commands then download the data, map the events, and train and score the models. All inputs other than the Sentinel-1 stage come from global open datasets, and that stage needs only a noncommercial Earth Engine project. The parts that remain specific to Bangladesh are the event lists and the asset weights, both of which are set in the configuration files and not in the code. The input data and outputs of the three validation regions and the landslide region are archived with checksums and licences [17].

5. Limitations and conclusions

Four limitations bound what the software can currently deliver. Exposure depends on OpenStreetMap, whose completeness varies widely [18] and has not been measured for rural Bangladesh, so composite risk describes mapped

exposure. Vulnerability uses population density and three access distances and leaves out income, housing and age structure. The terrain hazard surface is close to or below chance wherever embankments, tides or flash floods decide where water lies, from the south-west coast to the Jamuna east bank, and a model there would need embankment and tide data the pipeline does not yet ingest. Regions with few mapped floods, such as West-central, need more events before their scores can be ranked with confidence. Retraining the landslide model on the multi-year inventory of Rabby and Li [19] is the most direct improvement to the fourth region.

Fermium Hazard Mapper joins radar-observed flood labels, spatially blocked validation and a public dashboard in one open and reproducible chain. Its main contribution is a validation suite that runs every comparison on identical held-out areas and reports it with its spread, together with a design that stops on the kinds of silent error that invalidated its own first release.

CRedit authorship contribution statement

Md Rakib Hasan: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Shoumik Zubyer:** Conceptualization, Writing – review & editing.

Declaration of competing interest

Md Rakib Hasan is affiliated with Fermium Systems, which hosts the public dashboard. The software is released under the MIT licence and the dashboard is free to use. The authors declare no other competing interests.

Data availability

The code is at <https://github.com/rakibhhridoy/rimes-mapathon>, and the release this article describes is archived at **[software DOI, assigned on release]**. Input data and outputs for the three validation regions and the landslide region are archived at <https://doi.org/10.5281/zenodo.22978729>, and the national partition is reproduced by the configuration files in the repository.

Acknowledgements

Infrastructure data are © OpenStreetMap contributors, under the Open Database Licence. Population data are from WorldPop, University of Southampton, CC BY 4.0. Administrative boundaries are from geoBoundaries, William & Mary geoLab, CC BY 4.0. Sentinel-1 data are Copernicus Sentinel data, processed in Google Earth Engine. Landslide locations are from NASA's Co-operative Open Online Landslide Repository. The scenario maps shown beside the results are the Department of Disaster Management's Multi-hazard Risk and Vulnerability Assessment, served by the United Nations Satellite Centre (UNOSAT), and the basemap relief is from Esri.

References

- [1] A. R. M. T. Islam, S. Talukdar, S. Mahato, S. Kundu, K. U. Eibek, Q. B. Pham, A. Kuriqi, N. T. T. Linh, Flood susceptibility modelling using advanced ensemble machine learning models, *Geoscience Frontiers* 12 (2021) 101075. doi:10.1016/j.gsf.2020.09.006.
- [2] S. Talukdar, B. Ghose, Shahfahad, R. Salam, S. Mahato, Q. B. Pham, N. T. T. Linh, R. Costache, M. Avand, Flood susceptibility modeling in Teesta River basin, Bangladesh using novel ensembles of bagging algorithms, *Stochastic Environmental Research and Risk Assessment* 34 (2020) 2277–2300. doi:10.1007/s00477-020-01862-5.
- [3] R. Islam, P. Chowdhury, Local-scale flash flood susceptibility assessment in northeastern Bangladesh using machine learning algorithms, *Environmental Challenges* 14 (2024) 100833. doi:10.1016/j.envc.2023.100833.
- [4] M. S. G. Adnan, Z. S. Siam, I. Kabir, Z. Kabir, M. R. Ahmed, Q. K. Hassan, R. M. Rahman, A. Dewan, A novel framework for addressing uncertainties in machine learning-based geospatial approaches for flood prediction, *Journal of Environmental Management* 326 (2023) 116813. doi:10.1016/j.jenvman.2022.116813.
- [5] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillerá-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, C. F. Dormann, Cross-validation

strategies for data with temporal, spatial, hierarchical, or phylogenetic structure, *Ecography* 40 (2017) 913–929. doi:10.1111/ecog.02881.

- [6] P. Ploton, F. Mortier, M. Réjou-Méchain, N. Barbier, N. Picard, V. Rossi, C. Dormann, G. Cornu, G. Viennois, N. Bayol, A. Lyapustin, S. Gourlet-Fleury, R. Pélissier, Spatial validation reveals poor predictive performance of large-scale ecological mapping models, *Nature Communications* 11 (2020) 4540. doi:10.1038/s41467-020-18321-y.
- [7] A. Twele, W. Cao, S. Plank, S. Martinis, Sentinel-1-based flood mapping: a fully automated processing chain, *International Journal of Remote Sensing* 37 (2016) 2990–3004. doi:10.1080/01431161.2016.1192304.
- [8] K. Uddin, M. A. Matin, F. J. Meyer, Operational Flood Mapping Using Multi-Temporal Sentinel-1 SAR Images: A Case Study from Bangladesh, *Remote Sensing* 11 (2019) 1581. doi:10.3390/rs11131581.
- [9] UN-SPIDER, Recommended Practice: Flood Mapping and Damage Assessment Using Sentinel-1 SAR Data in Google Earth Engine, <https://un-spider.org/advisory-support/recommended-practices/recommended-practice-google-earth-engine-flood-mapping>, 2019.
- [10] J.-F. Pekel, A. Cottam, N. Gorelick, A. S. Belward, High-resolution mapping of global surface water and its long-term changes, *Nature* 540 (2016) 418–422. doi:10.1038/nature20584.
- [11] W. L. Hamilton, R. Ying, J. Leskovec, Inductive Representation Learning on Large Graphs, *Advances in Neural Information Processing Systems* 30; arXiv:1706.02216, 2017. arXiv:1706.02216.
- [12] B. Zadrozny, C. Elkan, Transforming classifier scores into accurate multiclass probability estimates, in: *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining*, 2002, pp. 694–699. doi:10.1145/775047.775151.
- [13] European Commission Joint Research Centre, Index for Risk Management – INFORM: Concept and Methodology, Version 2017, Technical Report, Publications Office of the European Union, 2017. doi:10.2760/094023.

- [14] A. Getis, J. K. Ord, The Analysis of Spatial Association by Use of Distance Statistics, *Geographical Analysis* 24 (1992) 189–206. doi:10.1111/j.1538-4632.1992.tb00261.x.
- [15] M. Barbet-Massin, F. Jiguet, C. H. Albert, W. Thuiller, Selecting pseudo-absences for species distribution models: how, where and how many?, *Methods in Ecology and Evolution* 3 (2012) 327–338. doi:10.1111/j.2041-210x.2011.00172.x.
- [16] B. Tellman, J. A. Sullivan, C. Kuhn, A. J. Kettner, C. S. Doyle, G. R. Brakenridge, T. A. Erickson, D. A. Slayback, Satellite imaging reveals increased proportion of population exposed to floods, *Nature* 596 (2021) 80–86. doi:10.1038/s41586-021-03695-w.
- [17] M. R. Hasan, Fermium hazard mapper: input data and outputs for flood and landslide susceptibility in Bangladesh (version 2), 2026. doi:10.5281/zenodo.22978729.
- [18] B. Herfort, S. Lautenbach, J. Porto de Albuquerque, J. Anderson, A. Zipf, A spatio-temporal analysis investigating completeness and inequalities of global urban building data in OpenStreetMap, *Nature Communications* 14 (2023) 3985. doi:10.1038/s41467-023-39698-6.
- [19] Y. W. Rabby, Y. Li, Landslide Inventory (2001–2017) of Chittagong Hilly Areas, Bangladesh, *Data* 5 (2019) 4. doi:10.3390/data5010004.